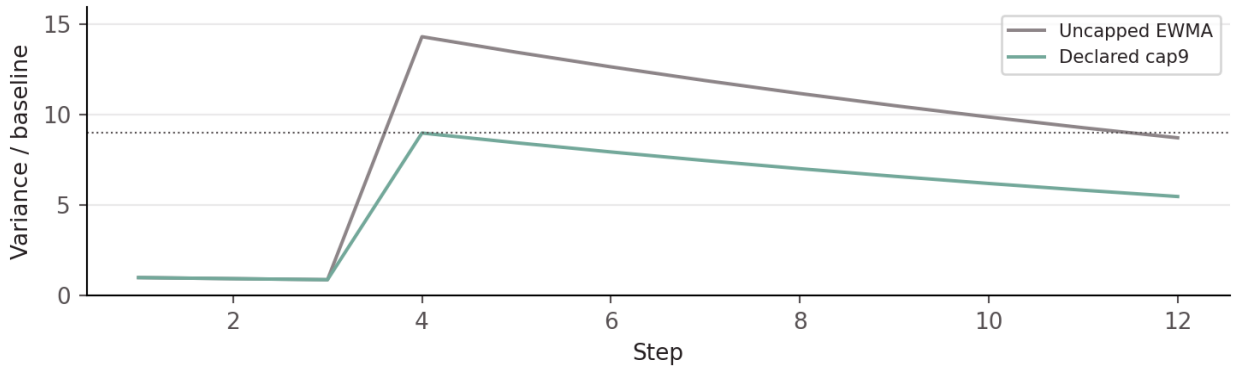
Changing volatility, finite price moments. Still research.

The annual conditional-volatility extension is built and tested. It fixes the mathematical price-moment blocker through an explicit variance ceiling. Weekly forecasts improve against iid, but annual scores are mixed and the portfolio evidence does not show a better strategy. The current default and funding limits are unchanged.

What changed

Each asset and FX variance updates from the previous shock. The variance state stays between 0.05 and 9 times the past baseline variance. The ceiling is three times the baseline standard deviation, not a 9x return limit. Decay stays 0.94. These are declared research settings, not fitted market truths or an ordinary GARCH model.



Illustrative hand fixture, NOT market data: a 15-baseline-SD shock at step3. The cap changes later volatility; it does not trim the shock or limit return/price support. Both paths use the same subsequent zero-shock fixture.

Means, shrunk correlation, common Gaussian-mixture states, cash and the property latent-factor assumption remain as before. The selected model works through the installed annual engine, optimizer, independent judge and cash ledger. It is opt-in research, not a new automatic dashboard choice.

Finite moments are proved, not inferred from finite simulations

A bounded conditional variance gives a finite upper bound on the conditional exponential moment of every finite listed/FX/property exposure. Repeated conditioning extends that bound to 52 weeks and any finite horizon. Returns stay unbounded. This is not the iid exact moment formula, a stationarity proof, a tail-calibration certificate or protection from floating-point overflow.

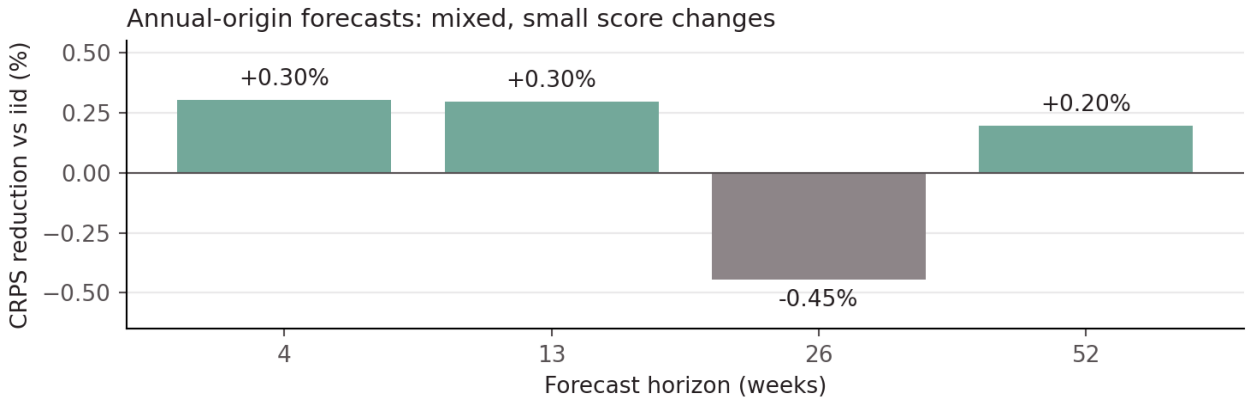
Weekly gain, modest annual evidence

For 2019-2025, mean normalized one-week CRPS fell from 0.675607 to 0.665920, a 1.43% reduction. Lower is better. Exact marginal scores, past-only annual refits and after-observation state updates were fixed before the run. This is not an investment return gain.

Model	Weekly CRPS	1% misses /1,825	5% misses /1,825
iid baseline	0.675607	18	71
Bounded cap9	0.665920	33	109
Cap4 sensitivity	0.665187	33	109
Cap16 sensitivity	0.665920	33	109

The primary paired normalized score difference is -0.009687; exploratory 20-week block 95% percentile interval [-0.015531, -0.003630]. Prespecified 10/40-week sensitivities also lie below zero. Dates were previously viewed, so this is a fixed retrospective comparison, not blind preregistration. Five assets are not independent weekly replications.

The 2008 development slice still has 31 /260 lower-5% misses (11.92%) under cap9, versus 56 /260 (21.54%) iid. That is improvement, not nominal 5% calibration. In the final slice the cap9 rate is 5.97%, and individual assets differ. Better general scores do not certify joint tails or funding.



2007-2025: 19 year-end origins, five listed/FX factors, 400 simulated paths at each origin for the endpoint score. Exact empirical CRPS of cumulative log returns, normalized by initial weekly SD times square-root horizon. Paired common random inputs; no cap selection from these scores.

At 52 weeks the average score improves only about 0.20%; at 26 weeks it worsens about 0.45%. At 52 weeks only emerging-market equity improves among the five asset-average point scores. There is no multihorizon or annual superiority claim.

A candidate must face more than its own model

The current-snapshot confirmation used 200 training paths and four batches each of 400 validation and 800 final paths. Both iid and bounded cap9 must accept a validation blend. All portfolios and blend trials were saved in a source/config-hashed lock before final scoring. Cap4, cap16 and FHS are extra final judges, not selection opportunities.

CARA portfolio	iid path-tail upper /6% limit	Bounded cap9 upper /6% limit
iid-trained raw	6.66% - FAIL	5.71% - PASS
Bounded-trained raw	8.44% - FAIL	6.79% - FAIL
iid-trained, 50% candidate /50% anchor	4.47% - PASS	4.06% - PASS
Bounded-trained, 25% candidate /75% anchor	4.02% - PASS	3.56% - PASS

The anchor is 90% cash and 10% fixed property. Blends are not merely “a smaller risk setting”: holdings actually change. The bounded CARA blend has about 74.0% cash, 12.4% IEF, 3.6% VTI and 10% property. These are frozen research candidates, not recommended holdings. Raw and blended portfolios are not interchangeable.

The two CARA blends pass all current final risk/holdings checks under iid, cap9, cap4, cap16 and FHS. Raw bounded CARA fails the path-loss limit under every final judge. No final failure changed a portfolio, threshold or model setting. A numerical failure of the first CARA formulation succeeded through the existing alternate formulation and stays in the evidence.

Funding budget is not payment execution

All current final paths pay the small default $3M annual schedule for the reported portfolios. That result is unsurprising beside $100M and says little about a demanding liquidity plan. Nominal and stressed cash-ledger summaries are retained separately from W1-buffered provisioning-budget checks.

Counterfactual schedule and portfolio	iid funded fraction	Bounded funded fraction
$95M final; 90% cash +10% illiquid property	0%	0%
$95M final; equal free holdings	54.09%	55.84%
$60M week4 +$30M final; equal free holdings	74.81%	81.44%
$60M week4 +$30M final; cash/property anchor	100%	100%

Stress fixtures, NOT user obligations. For the $95M final call the cash/property anchor PASSES the 20%-tail-shortfall funding budget while missing $5M on every path. Equal holdings also pass that funding budget despite many unpaid calls. These modeled fractions are not calibrated economic probabilities. No stricter funding rule was installed.

The annual failures stay visible

All 19 historical origins completed with no omitted years: 2006-12-29 through 2024-12-27, producing annual 2007-2025 outcomes. Listed and FX paths are actual historical data; property is a latest-vintage quarterly index carried forward under an assumed 45-day release lag. It is not a revision-safe tradable-price backtest.

The prespecified historical preview tier had only two validation/final batches. Its Bonferroni Student-t bounds were too imprecise to accept ANY validation blend or final portfolio, including the cash/property anchor. No historical portfolio is labelled approved. The failed tier was not rerun with looser risk limits or a hand-picked set of years.

For example, at the first origin the anchor's average reference path-loss tail is about 3.33%, but its simultaneous upper diagnostic exceeds 100%. That is uncertainty from two batches and 56 comparisons, not a forecast of losing more than all capital. It does not establish that the policy is unsafe or safe. More precision would need a new declared validation exercise, not deletion of this failure.

Raw, non-approved policy	Mean ledger net return	Worst year	Paid default in
iid-trained CARA	1.35%	-11.84%	19 /19 years
Bounded-trained CARA	1.25%	-16.21%	19 /19 years
Equal /cash reserve	3.99%	-14.98%	19 /19 years
Cash +fixed property	-2.38%	-3.98%	19 /19 years

Net ledger return includes the $3M withdrawn; it is not investment profit before payouts or a risk-adjusted comparison. Equal and cash-reserve coincide here because equal holdings already reserve more than $3M. Different risk exposures make the larger raw mean no evidence of superiority.

Bounded-minus-iid raw CARA return averages -0.103 percentage points, with individual annual differences from -5.29 to +4.96 points. Matched mean-variance averages -0.133 points. No annual calibrated comparison exists at this failed tier, so there is no manufactured mean for accepted portfolios.

Model limits and numerical limits are separate

All 76 historical CARA/matched-MV solves succeeded through the existing solver route. Eleven first-formulation failures remain recorded, plus numerical-inaccuracy warnings. Those successful solves did not pass the independent risk judge. Optimization success is not risk acceptance.

The cap9 historical update ceiling is hit seven times across the five assets in the 2008 slice, and zero times in final 2019-2025 scored updates. In current final simulations it is hit 122 times out of 832,000 stochastic-factor updates (0.015%); cap4 hits 1,228 and cap16 hits 21. Rare observed hits do not make the cap economically harmless: it still limits crisis persistence.

What this adds, and what remains open

Built and checked

341 tests pass in 15.54s, with three recorded warnings. An explicit opt-in bounded conditional joint-path law; a finite-horizon moment proof; hand recurrence and covariance/moment oracles; large-shock/cap/zero-cash/prefix tests; default-invariance and annual CLI integration checks; saved cross-model validation locks; state-hit diagnostics; weekly, multihorizon, current-funding and 19-year historical evidence. No new dependency, GPL code, public hosting, contact or spend.

The right conclusion

The extension is mathematically usable over a full year. It is not established as the best return model or a better investment policy. The weekly gain is more convincing than the annual gain; the cap is a visible assumption; crisis tails and cross-asset/property dependence remain imperfect. Keep the existing default unless you explicitly decide otherwise.

Highest-value next work

Your next priorities are the stricter all-payments funding mode, this bounded annual model, and past-only W1-radius calibration. Integrating the optimizer with executable cash flows is parked for now. Costs, settlement and joint crash/property evidence remain open. No rule can force a feasible99% outcome if the obligations exceed what the allowed assets can fund.

Source and reproducibility notes

User-supplied Meucci (2005), Risk and Asset Allocation. Printed pp.103-105,123-125,233-234. Projection/invariance context; no pages redistributed.

RiskMetrics (1996): EWMA method context. The cap and weekly setting here are our declared variant, not its calibrated daily recommendation. Source.

Gneiting & Raftery (2007): proper CRPS, empirical distributions. Source.

Patton (2011): QLIKE and noisy-proxy qualifications. Source.

Gneiting, Balabdaoui & Raftery (2007): calibration/sharpness limits. Source.

Official arch documentation: conditional forecast timing. No arch dependency added. Source.

Pritsker /Federal Reserve (2001): historical-simulation and FHS limitations. Source.

Protocol and moment proof were committed before scientific runs. Pilot 1.99s, current confirmation 8.74s, all historical origins 44.47s. Every final score is tied to source/data/config evidence; post-output changes only improve labeling and integration checks. Private evidence includes raw results, failures, locks and hashes.